

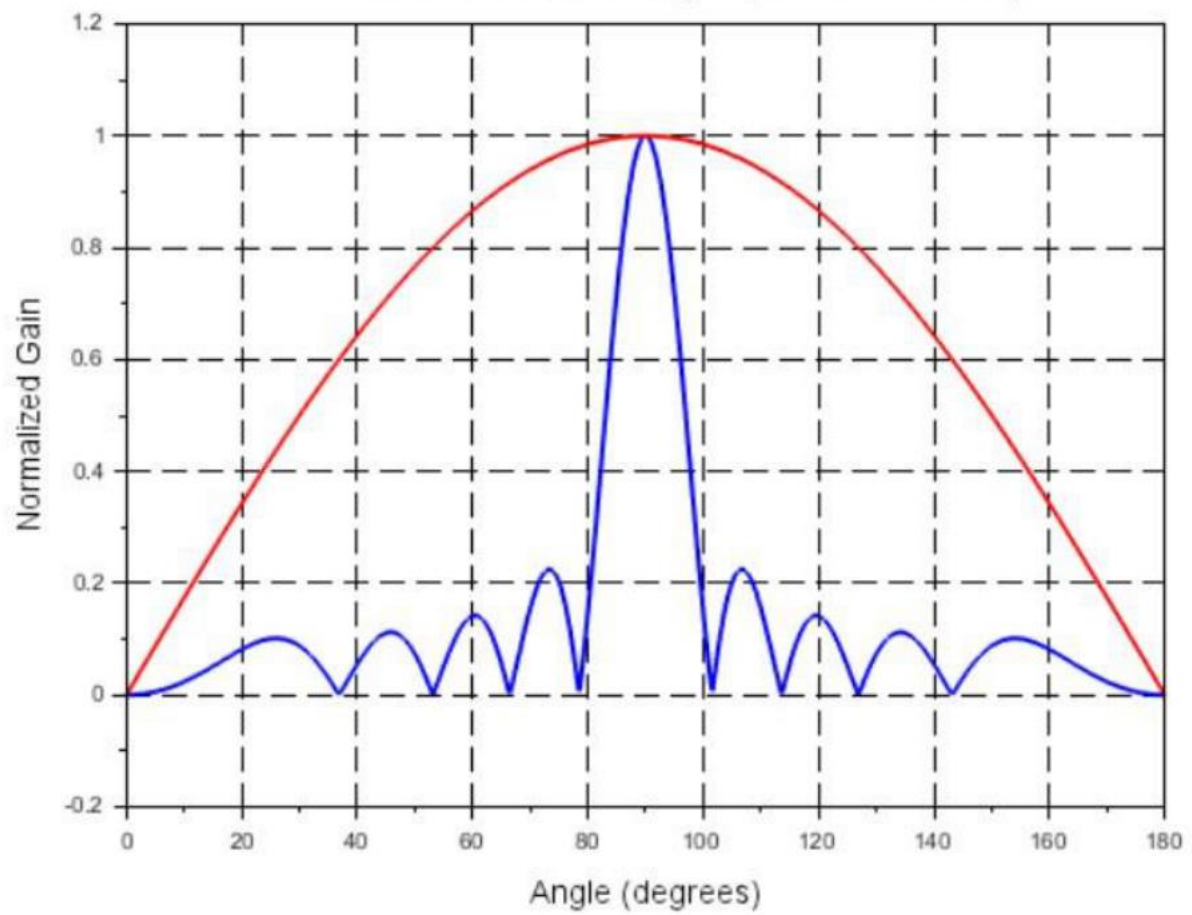
EXP:17

Determination and plotting of Antenna Gain Calculation.

```
1  clc;
2  clear;
3  N = 10; .....//Number of elements
4  d_lambda = 0.5; .....//Distance between elements in terms of wavelength
5  beta = 0; .....//Phase shift
6  theta_deg = linspace(0, 180, 1000);
7  theta = theta_deg * (%pi / 180); .....//Convert to radians
8  psi = 2 * %pi * d_lambda * cos(theta) + beta;
9  AF = zeros(1, length(psi));
10 for k = 1:length(psi)
11     if abs(sin(psi(k)/2)) < 1e-7 then
12         AF(k) = N; .....//Main beam peak limit
13     else
14         AF(k) = sin(N * psi(k) / 2) / (N * sin(psi(k) / 2));
15     end
16 end
17 AF_normalized = abs(AF) / max(abs(AF));
18 scf(0); .....//Open graphic window 1
19 clf();
20 plot(theta_deg, AF_normalized, 'thickness', 2, 'color', 'blue');
21 title("Array Factor vs Theta", "fontsize", 4);
22 xlabel("Theta (degrees)", "fontsize", 3);
23 ylabel("Normalized Array Factor", "fontsize", 3);
24 ax = gca();
25 ax.grid = [1, 1]; .....//Turn on grid for x and y
26 ax.grid_style = [2, 2]; .....//Dashed style
27 set(ax, "data_bounds", [0, -0.05; 180, 1.05]);
28 scf(1); .....//Open graphic window 2
29 clf();
30 rho = abs(sin(theta)); .....//Normalized gain or pattern example
31 polarplot(theta, rho);
32 title("Polar Radiation Pattern", "fontsize", 4);
33
```

OUTPUT:

Normalized Gain vs Angle (Cartesian Plot)



Polar Radiation Pattern

